

Chapter 2: Heat & Thermodynamics

Temperature Scales, Laws of Thermodynamics, Heat Transfer & Entropy

2.1 Temperature & Thermal Expansion

Temperature is a measure of the average kinetic energy of the particles in a substance. Conversions between primary temperature scales are given by:

- Celsius to Fahrenheit: $(F = \frac{9}{5}C + 32)$
- Celsius to Kelvin: $(K = C + 273.15)$

Thermal expansion accounts for the change in dimensions of materials due to temperature variation: linear expansion $(\Delta L = \alpha L_0 \Delta T)$ and volumetric expansion $(\Delta V = \beta V_0 \Delta T)$.

2.2 Laws of Thermodynamics

Thermodynamics governs the rules of energy transformation in physical systems:

- **Zeroth Law:** Establishes thermal equilibrium and temperature as an empirical property.
- **First Law (Conservation of Energy):** The change in internal energy of a system (ΔU) is equal to the heat added to the system (Q) minus the work done by the system (W) : $(\Delta U = Q - W)$.
- **Second Law:** Entropy of an isolated system always increases over time; heat cannot spontaneously flow from a colder body to a hotter body.
- **Third Law:** As temperature approaches absolute zero, the entropy of a system approaches a constant minimum.

2.3 Modes of Heat Transfer

Heat transfers via three distinct mechanisms:

1. **Conduction:** Direct microscopic exchange of kinetic energy between particles through solid mediums (Fourier's Law: $(q = -k \nabla T)$).
2. **Convection:** Bulk movement of fluid molecules within liquids and gases.
3. **Radiation:** Emission of electromagnetic energy from the surface of objects (Stefan-Boltzmann Law: $(P = \sigma A T^4)$).